CODECAFE SIGNALS

AI Trading Copilot & Practice Platform

Project report: what was built, how it was tested, results, and how to run it

Period: 3 - 6 November 2026 | Status: working prototype, ready for client demos

Headline	Result
Automated test suites	5 of 5 suites pass; full-system suite 17 of 17 checks pass (re-run 6 Nov 2026)
Live demo trading (MT5, broker-verified)	4 automated trades, 3 wins / 1 loss, net +$138.07 on a $100,000 demo account
Strategy research (forex, unseen data)	+107R over 445 trades, profit factor 1.35, 12 of 16 markets profitable
AI screen reading	Reads a chart screenshot from any platform and returns BUY / SELL / WAIT with entry, SL, TP
Multi-client web platform	Logins, admin-created client accounts, separate paper account per client
Cost to run the demo	Rs 0 - free Gemini AI tier, free hosting plan (Hugging Face + Supabase)

All profit figures come from demo or paper (fake-money) accounts. No real money was used. Results are a small sample and are not a promise of future returns - see section 7.

1. What was built

CodeCafe Signals is a trading assistant with three layers: it **reads the market** (live prices, news, chart structure), it **proposes trades** with mandatory stop-loss and position sizing, and it **executes only on demo or paper accounts**, after approval or within hard risk limits.

Component	What it does	Main files
Web dashboard	Login-protected site: live quotes, news sentiment, AI signal, backtest, open positions, kill switch	dashboard_server.py, web/index.html
Screen Coach (AI)	User shares any broker screen (Zerodha, Bybit, Binance, MT5, TradingView). AI reads it and gives a plan. No broker password or API key needed	coach.py
Client accounts	Admin creates client logins; clients change their password; shared demo login for pitches	auth.py, web/login.html
Paper broker	Fake money at real prices for crypto, forex, US and Indian stocks; real candles settle SL/TP; max 2% risk per trade; separate account per client	paper.py, store.py
MT5 forex engine	Live MetaTrader 5 demo trading with approval gate and hard limits	server.py, auto_trader.py
Crypto engine	Bybit market data + demo-only order client, long-only spot	bybit.py, crypto_server.py
Strategy research	Strategy v2 (EMA cross + RSI + trend + session filters), backtester, variant search	signals.py, research.py
Market intelligence	News scraping + sentiment, economic calendar, regime, market structure, ML model	news_scraper.py, intelligence.py, regime.py, ml_pipeline.py
Claude Code tools	3 MCP servers so the AI assistant can scan, backtest and trade (with approval) from chat	server.py, crypto_server.py, omni_server.py
Cloud packaging	Docker image + free hosting plan (Hugging Face Spaces + Supabase Postgres)	Dockerfile, README.md

Safety rules enforced in code

- Every order needs a stop-loss; orders without one are refused.
- Risk per trade capped (1% on MT5, 2% on paper); MT5 lot size capped at 0.10; crypto order size capped at 1,000 USDT.
- Daily loss circuit breaker at 3%; at most 3 open positions.
- Real-money endpoints are blocked: MT5 refuses live accounts, Bybit client only talks to Demo/Testnet.
- Web site places paper trades only - clients place real orders themselves.

2. Test results

Automated test suites (re-run 6 November 2026)

Suite	What it checks	Result
test_core.py	Indicators, strategy v2 signals, backtester costs, long-only crypto mode	**PASS**
test_omni.py	Chart pattern detection, support/resistance, news sentiment, paper broker (fills, SL/TP, 2% risk cap, per-user accounts)	**PASS**
test_auth.py	Passwords, wrong-password refusal, tampered session cookie, password change, last-admin protection	**PASS**
test_prototype.py	Watchlist, database, paper broker, live gold price, Groq news AI	**PASS**
test_full_system.py	17 end-to-end checks (data, structure, regime, calendar, AI news, strategy, backtest, risk gate, broker, trailing stop, kill switch, REST API with login, ML, knowledge base, reconciliation, streaming)	**PASS 17/17**

Live tests performed during development

Test	Outcome	Status
MT5 demo connection + 0.01 lot EURUSD order opened and closed	Order executed on MetaQuotes-Demo, closed at $0.00	**PASS**
Automated trader on MT5 demo (4 Nov)	4 trades placed and closed by the bot (details in section 3)	**PASS**
Real prices for paper trading	RELIANCE.NS 1311, BTCUSDT 78,542, EURUSD 1.1257 fetched live	**PASS**
AI screen reading (Gemini) on a live BTC chart	Correctly described an 81,000 to 75,500 drop and sideways range; answered WAIT with reasons	**PASS**
Login security in the browser	Anonymous access refused (401); wrong password refused; client blocked from admin page (403); demo login cannot change password	**PASS**
Separate client accounts	Client A sees 1 position, client B sees 0 for the same server	**PASS**
Site speed under load	Fixed a freeze: health check went from 35 s to under 0.1 s while a dashboard loads	**PASS**
Bybit demo API keys	Key was rejected by Bybit (401); India blocks Bybit testnet (error 10024)	**OPEN**
Docker cloud image build	Not yet run (Docker Desktop was off); Hugging Face builds it on upload	**OPEN**

3. Profit & loss

Automated trading on MetaTrader 5 demo (verified from broker history)

Account: MetaQuotes-Demo (demo account confirmed by the broker API), starting balance $100,000. The figures below were read directly from MT5's deal history on 6 Nov 2026, and match the bot's own journal.

Time (UTC)	Symbol	Side	Lots	Entry	Exit	Net P&L	Result
4 Nov 10:24	XAUUSD (Gold)	BUY	0.10	4033.43	4046.66	+$132.30	**WIN**
4 Nov 10:25	EURUSD	BUY	0.10	1.12381	1.12426	+$4.50	**WIN**
4 Nov 10:26	AUDUSD	BUY	0.10	0.69998	0.70083	+$8.50	**WIN**
4 Nov 12:36	USDCAD	SELL	0.10	1.39547	1.39648	-$7.23	**LOSS**
Total						+$138.07	**3W / 1L**

Other activity on the same account: one 0.01 lot test trade ($0.00) and three 0.01 lot manual GBPUSD trades (-$0.08). Account balance after all trades: **$100,137.99**.

Honest reading: 4 trades is far too few to prove an edge. One gold trade made 96% of the profit.

Strategy research (historical backtests)

Method: 32 strategy variants were tested on the first 70% of history; the winner (v2) was then checked once on the last 30% it had never seen. 1R = the amount risked on one trade; spreads and fees included.

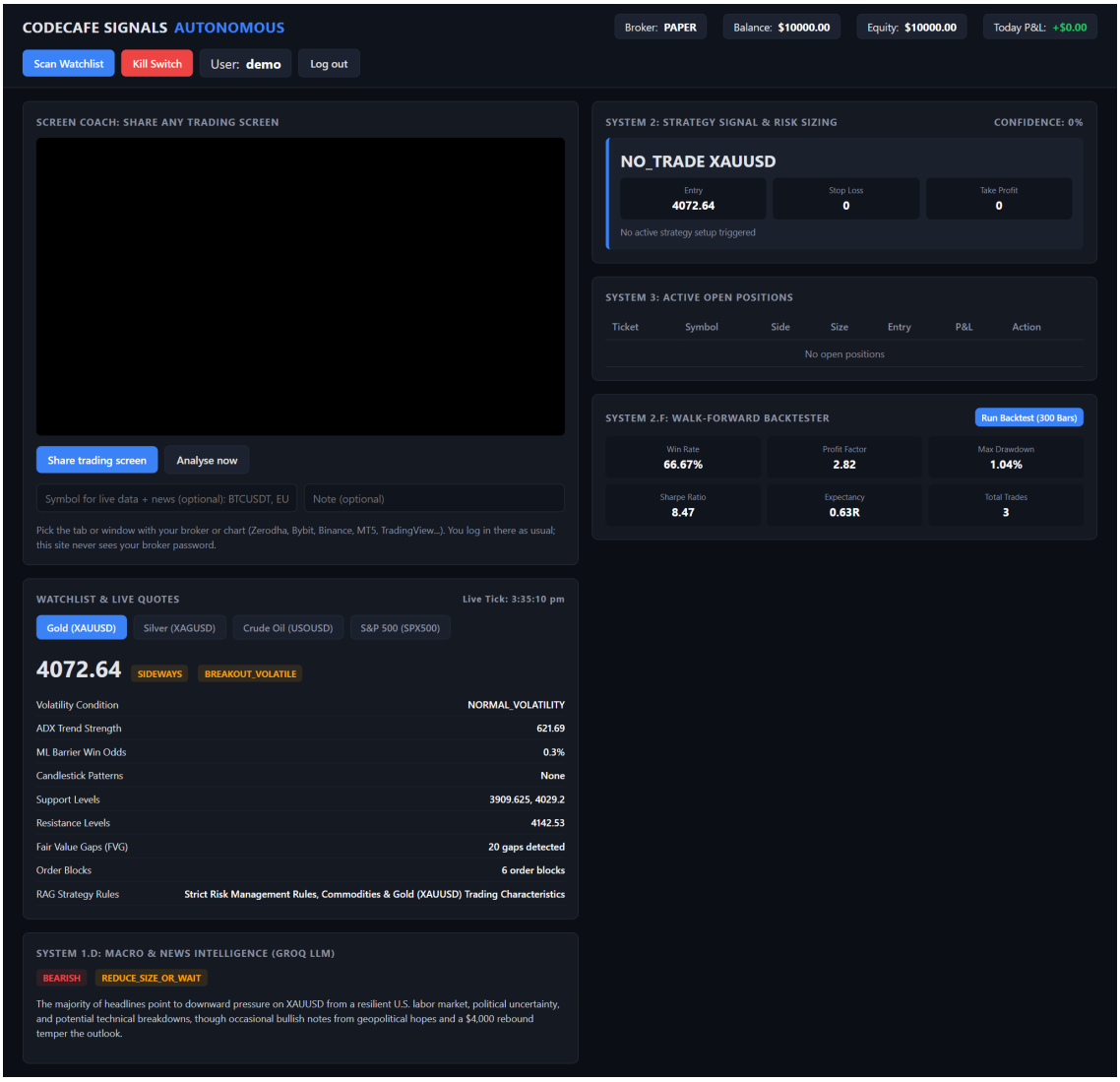
Test	Trades	Result	Profit factor	Verdict
v1: EMA cross + RSI, forex, 4 pairs, 8-16 years	-	about breakeven; -131R on last 30%	0.94 - 1.09	Rejected
v2 forex, unseen last 30%, 8 pairs x H1/H4	445	+107R, win rate 31%, 12 of 16 markets up	1.35	Best so far
v2 full H1 history: EURUSD / USDJPY / XAUUSD	-	+16R / +38R / +25R	-	Positive
v2 on crypto (H1, unseen 30%, 0.2% fees): BTC	19	+8.4R	-	Mixed
v2 on crypto: ETH / SOL / XRP	-	-14.5R / -13.0R / -14.1R	-	Loses
v2 BTCUSDT H1, full 2024-07 to 2026-11	31	+18.22R	1.90	Small sample

Conclusion: strategy v2 shows a modest edge on forex in research; on crypto it does not hold up except BTC. A crypto-specific strategy is still needed.

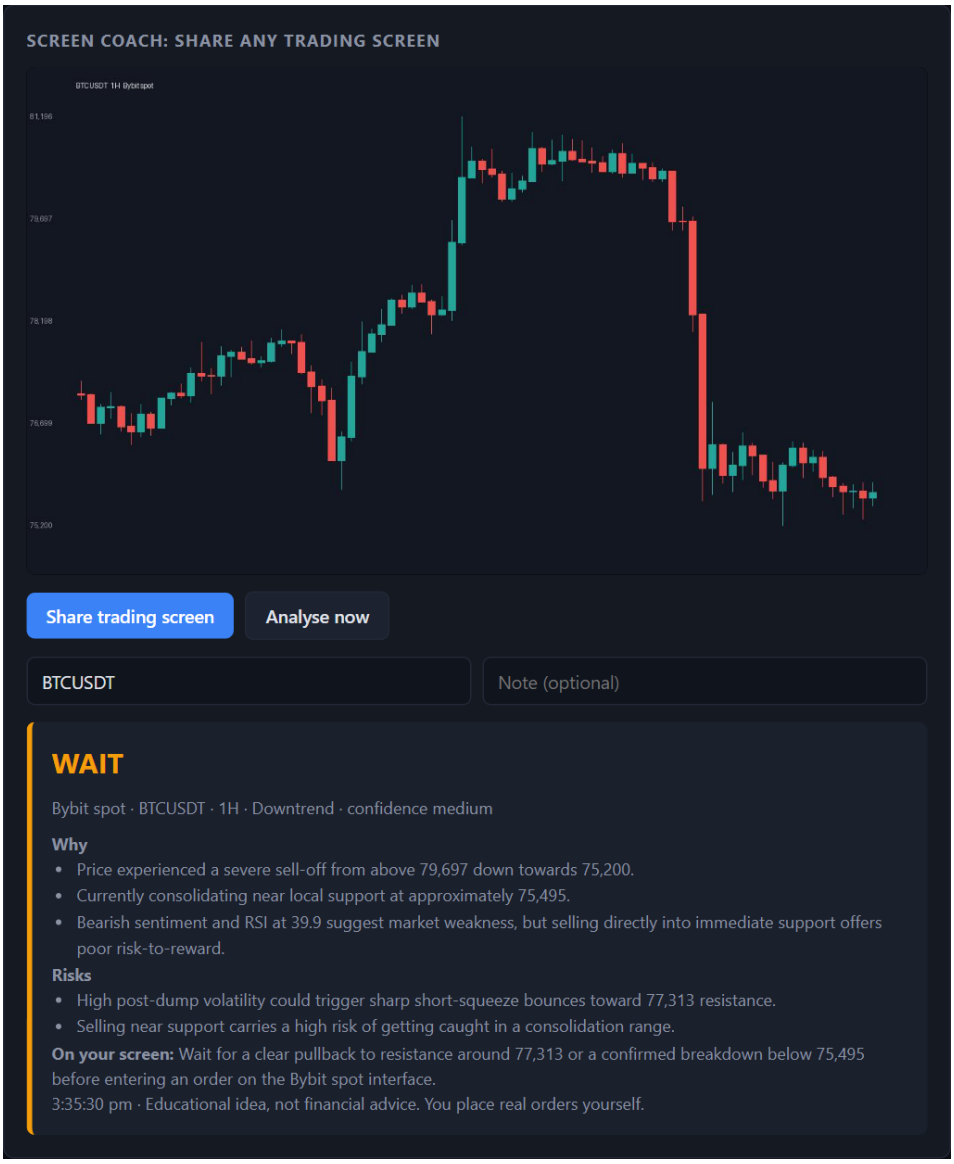
Paper trading

Paper accounts start with $10,000 and Rs 10,00,000 of fake money per user. They were used for functional testing (fills, stop-loss, take-profit, closing), not for performance measurement, so their P&L is not reported as a result.

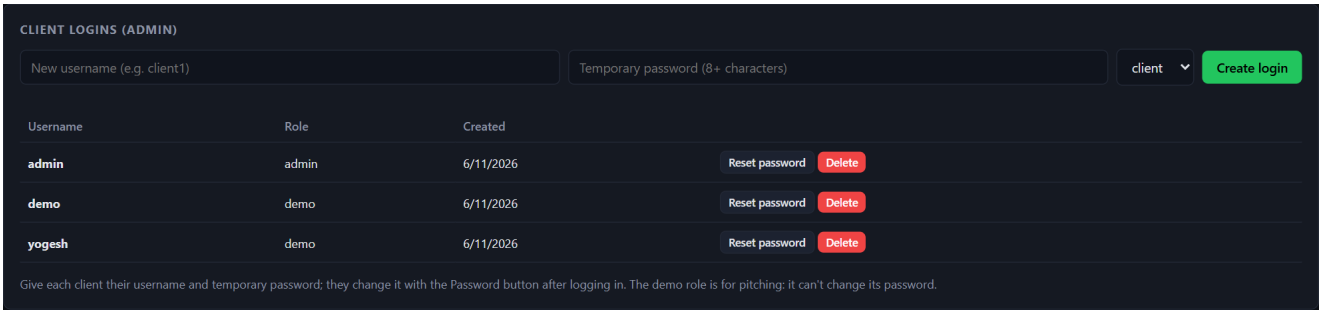
4. The product (screenshots of the running app)



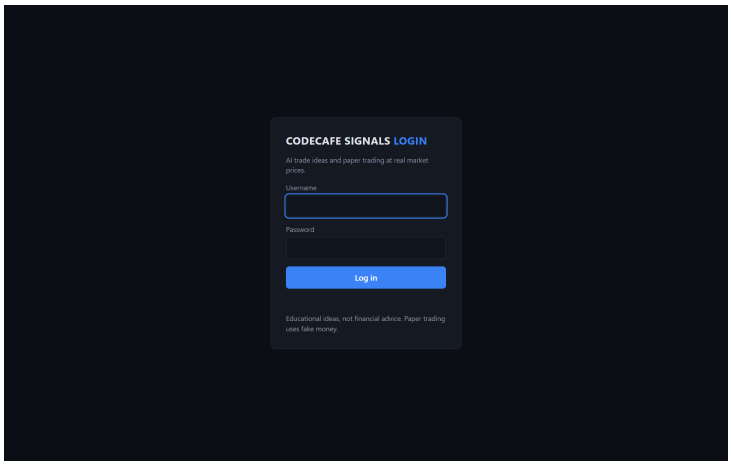
Dashboard after logging in with the demo account: live gold price, strategy signal (here NO_TRADE: no valid setup at that moment - it only signals when rules agree), news intelligence, backtester and open positions.



Screen Coach: the AI read a BTCUSDT 1-hour chart (frame as sent by screen share), combined it with live RSI and news, and returned a plan with entry, stop-loss, take-profit, reasons and risks.



Admin panel: create a login for each client, reset passwords, delete logins.



Login page - every page of the site requires a login.

5. How to start

Run on a laptop (for demos today)

- Open the folder **O:\Projects\CodeCafe Signals** and double-click **Run_CodeCafe_Signals.bat**.
- Wait 30-60 seconds; the browser opens **http://127.0.0.1:8780**.
- Log in with the demo account (username and password are in the project's .env file: DEMO_USER / DEMO_PASSWORD).

Demo script (about 5 minutes)

- **Login page** - each client gets a private login.
- **Dashboard** - live prices, news sentiment, AI signal with entry, stop-loss and take-profit.
- **Screen Coach** - click Share trading screen, choose a chart tab (TradingView, Zerodha, Bybit...), click Analyse now. The plan appears in 10-30 seconds.
- **Paper trade this** - place the idea with fake money; show it in Open Positions, then Close.
- **Run Backtest** - the strategy checked on history.
- **Admin** - log in as admin, click Clients, create a client login live.

Put it online for free

- Supabase (free): create a project, copy the Session pooler connection string.
- Google AI Studio (free): create a Gemini API key.
- Hugging Face: create a Docker Space; add the keys and admin/demo logins as Secrets; upload the project (never the .env file).
- Share the Space link with clients; create a login for each from the admin panel.

Developer commands

```
uv run python test_core.py
uv run python test_omni.py
uv run python test_auth.py
uv run python test_full_system.py
uv run python dashboard_server.py
```

6. Problems found and fixed

Problem	Fix
Bybit testnet blocks Indian accounts (error 10024)	Switched the crypto engine to Bybit Demo Trading; added a no-key paper broker so crypto practice works without any exchange account
Live demo data needed without paid APIs	Free sources: Yahoo Finance, Bybit public data, Google/Yahoo RSS news
Paid AI was not wanted for the demo	Free Gemini tier for screen reading, with automatic fallback to faster models when one is busy (the newest model hung for over 4 minutes in testing)
Whole site froze for 35 s while one user loaded the dashboard	Moved slow market-data work off the main server loop; now under 0.1 s
One shared account for everyone	Per-client logins and per-client paper accounts, verified isolated
Port 8765 blocked by a stuck Windows socket	App moved to port 8780 (configurable)
UI trade path claimed EXECUTED when it had only focused a window	Now reports honestly that no order was placed

7. Known limitations and next steps

- **No proven profitability yet.** 4 live demo trades and research backtests are encouraging but not proof. Next: run the strategy on demo/paper for 4-8 weeks and report the real track record.
- **Dashboard indicator bug:** the ADX value shown (e.g. 623) is outside its valid 0-100 range - to be fixed in regime.py. The dashboard's quick backtest uses only 300 bars (3 trades) - too short to judge.
- **Kill switch is global:** one user pressing it pauses trading for all users until restart.
- **Watchlist is fixed** to Gold, Silver, Crude Oil and S&P 500 on the dashboard (Screen Coach accepts any symbol).
- **Cloud deployment not yet done:** the Docker image has not been built; needs Supabase + Hugging Face accounts.
- **Free AI tier** has daily limits and can be slow; Google may use free-tier data to improve its models.
- **AI answers can vary:** on the same chart the Screen Coach answered WAIT in one run and SELL in another. Next: ask for a WAIT unless several signals agree, and log every answer to measure consistency.
- **Regulation:** giving buy/sell tips to paying members of the public in India requires SEBI Research Analyst registration; offshore forex with real money is not allowed under FEMA. The product is positioned as an education / practice copilot until then.